Scalar decays and LFV Higgs rates in the S_3-symmetric 3HDM: a gauge-phobic state for every δ, and an LFV bound on the lepton sector rather than the scalars

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Abstract

I have turned the LFV couplings of our draft [1] into branching ratios, over every viable point of the scalar scan. The CP-even basis $R_S = R_A R_H(\delta)$ is kept at *general* δ, so the draft's scenarios A and B appear only as limits. Three structural results follow. First, h_0 is gauge-phobic *exactly, for every* δ: $g_{h_0VV} = 0$ identically, while its LFV couplings do not depend on δ at all. Second, δ is not a free dial: the quartics predict it at each point, and $|\delta| \leq 0.32$ over the viable set. Third, the draft's Scenario-C relations, printed with "$\approx$", are exact. Phenomenologically, $\mathcal{B}(h \to \mu\tau)$ from the SM-like state exceeds the CMS limit [3] in 18% of entries, and that fraction is carried almost entirely by the lepton-sector parameter μ_3^ℓ. LFV data therefore bound μ_3^ℓ, not the scalar sector. Each CP-even state has its own LFV channels: the SM-like state never couples to the electron ($\tau\mu$ only), while the gauge-phobic h_0 has an electron on every coupling ($e\tau$, $e\mu$ only). This is the same reason that exact S_3 forces $V_{us} = 0$. (A first version of this report had h_0 decaying to $\tau\mu$ and no $e\tau$ anywhere. That came from a lepton Yukawa split with $G_2 = 0$, corrected in §6.) Two library gaps were closed along the way: VSS decays and the spin-0 $\gamma\gamma$ form factor.

1 Summary

Everything below comes from `research/thdm_s3/03_scalar_decays.ipynb`. It registers the physical basis on the `feynlag` model, extracts every Feynman rule from the Lagrangian, and runs the decay engine over the 1973 viable points of the scalar scan.

1. h_0 **is gauge-phobic for any** δ, and the other two CP-even states share the SM coupling as $\cos\delta$, $\sin\delta$ (§3).
2. δ **is predicted, and** $\delta = -\psi$, not the $\alpha - \theta_v$ I first used. That earlier formula mis-assigned the couplings; it is corrected here and in every downstream number (§4).
3. **The draft's Scenario-C relations are exact**, and Q_2 (the h_0 couplings) does not depend on δ (§5).
4. **LFV rates**: $h \to \mu\tau$ constrains μ_3^ℓ. The SM-like state never couples to the electron, and h_0 couples *only* through the electron ($e\tau$, $e\mu$; §6). This is the main phenomenological result, and it **corrects** the first version of this report (§6).
5. **Two library gaps closed**: VSS decays ($A \to Zh$, $H^\pm \to W^\pm h$) and the spin-0 form factor A_0 for charged-Higgs $\gamma\gamma$ loops (§7).

2 The physical basis

Five rotations are registered on the model: the three scalar rotations ($R_S = R_A R_H(\delta)$ for the CP-even states, R_A for the CP-odd and charged), plus the Weinberg and charged-current rotations. Every vertex below is therefore extracted from the Lagrangian in the physical basis rather than written down. $R_H(\delta)$ acts only in the 1–3 plane, and R_S is orthogonal for any δ.

3 The gauge couplings at general δ

R_A's first column is the unit vacuum direction $\hat{n}$ [1]. The hVV couplings come out as

$$g_{h_i VV} = g^{\rm SM}\left(R_S^{\mathsf{T}}\hat{n}\right)_i = g^{\rm SM}\,(\cos\delta,\ 0,\ \sin\delta), \qquad \sum_i g_{h_i VV}^2 = \left(g^{\rm SM}\right)^2,$$

for W and Z alike. The SM reference $g^{\rm SM} = i g_w m_W$ is read off the $\delta \to 0$ limit, not typed in. The middle zero is h_0: R_H never touches it, so its column of R_S is R_A's middle column, which is orthogonal to $\hat{n}$. The zero survives at a generic vacuum angle, so it is orthogonality, not the $\sqrt{3}$ alignment. [2]'s scenarios are the limits:

scenario	$g_{h_1VV}/g^{\rm SM}$	$g_{h_2VV}/g^{\rm SM}$	g_{h_0VV}
A ($\delta \to 0$)	1	0	0
B ($\delta \to \pi/2$)	0	1	0
C (general)	$\cos\delta$	$\sin\delta$	0

4 δ is predicted, and it is $-\psi$

δ is fixed by the residual CP-even 2×2 block, whose angle ψ the scalar-sector code computes at every point. **A correction I owe you here:** I first used $\delta = \alpha - \theta_v$. The correct relation is $\delta = -\psi$, for two reasons.

- R_H's $(0,2)$ block is $\mathrm{rot}(\delta)$, while the library's 2×2 solver returns the angle that diagonalises the transpose, which accounts for the sign.
- $R_A(\phi,\theta_v)R_H(\delta) = R_A(\phi,\theta_v+\delta)$ identically. So the draft's total angle is $\alpha = \theta_v + \delta$, and there is no θ_v to subtract from a block angle that is already measured from the geometric basis.

The old formula left $R_S^{\mathsf{T}} M_S R_S$ off-diagonal at the 10^5 level and mis-assigned the couplings: at benchmark point 0 it gave $\cos^2\delta = 0.55$ where the true value is 0.995. Every number in this report uses $\delta = -\psi$, which is checked against the stored scan values to 4×10^{-6} (the λ rounding).

Over the viable set $\delta \in [-0.322, 0.321]$ and $\cos^2\delta \in [0.900, 1.000]$. Because $\psi = \frac{1}{2}\arctan(\cdot) \in (-\pi/4, \pi/4)$, $\cos^2\delta \geq \frac{1}{2}$ always. h_1 is therefore by convention the state with the larger VV coupling, and scenario B is a symbolic limit that the scan cannot reach. The SM-like state is H_2 at every listed benchmark point.

5 The draft's Scenario-C relations are exact

[1] prints the Scenario-C LFV couplings with "$\approx$". They hold identically for all δ, because Q_i is linear in the columns of $R_S = R_A R_H(\delta)$: $Q_1(C)$ and $Q_3(C)$ are exact rotations of $Q_1(A)$ and

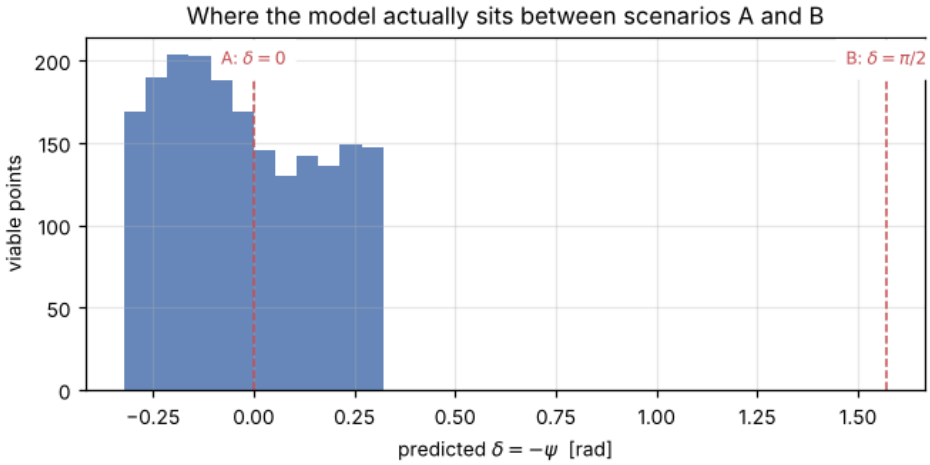


Figure 1: The predicted $\delta = -\psi$ over the 1973 viable points. The model sits near scenario A, and never near B.

$Q_3(A)$, and

$$Q_2(C) = Q_2(A) \qquad \text{exactly.}$$

Together with §3, this makes h_0 the state with no VV coupling and δ-independent lepton couplings. It is the one CP-even state whose LFV decays the alignment cannot suppress.

6 LFV rates against the CMS limits

With $\mathcal{L} \supset -h\,\bar{\ell}_a Q_{ab} P_R \ell_b + \text{h.c.}$, the width in the $m_\ell \ll m_h$ limit is

$$\Gamma(h \to \ell_a \bar{\ell}_b + \bar{\ell}_a \ell_b) = \frac{m_h}{8\pi}\left(|Q_{ab}|^2 + |Q_{ba}|^2\right),$$

which the engine reproduces from the extracted FFS vertices. Branching ratios are quoted against the SM prediction for the total width, $\Gamma_h = 4.1$ MeV [7]; the measured $3.7^{+1.9}_{-1.4}$ MeV [8] agrees but is far less precise. That is the standard convention in LFV searches, and the honest one here, since the quark Yukawas are not in this build.

The lepton Yukawas are fixed by the three masses up to one free parameter, μ_3^ℓ. The inversion is verified along the whole μ_3 range: the singular values of M_ℓ equal (m_e, m_μ, m_τ) to 10^{-9}, and a 1% change in μ_4 breaks it. The rates are evaluated at $\mu_3^\ell = 0.5,\ 0.9,\ 1.4$ GeV over all 1973 points, giving 35 514 (point, μ_3, state, channel) entries.

The per-doublet Yukawa matrices. $Q_i = \sum_j (R_S)_{ji} O^{\mathsf{T}} G_j O$ needs each doublet's Yukawa matrix G_k, not only $M_\ell = \sum_k v_k G_k$. The first version of this report wrote M_ℓ with the alignment's $\sqrt{3}$ typed in, which made the split return $G_2 = 0$. Q_1 and Q_3 only see the sum and were right; Q_2, the h_0 couplings, was wrong. The G_k are now read off the Lagrangian with all three VEVs symbolic. They agree with an independent computation in the draft's own basis to 10^{-17}, and the old split fails that comparison at the 10^{-2} level. The corrected Q_2 also equals the closed form $Q_2(A)$ printed in our draft [1] exactly. The draft had it right; only the first version of this calculation disagreed.

Which state couples to which flavours. In the eigenbasis of the doublet block, the vacuum direction $u = (\cos\varphi, \sin\varphi)$ projects as $(\cos\frac{3\varphi}{2}, \sin\frac{3\varphi}{2})$, which decouples the electron at $\varphi = \pi/3$.

The orthogonal direction (that of h_0) gives a reflection that anticommutes with the first, so it is purely off-diagonal and pairs the electron with everything else. Hence $h_{1,2}$ give $\tau\mu$ only and h_0 gives $e\tau$, $e\mu$ only, for any μ_3, δ and λ's. The scan confirms it at every entry.

state	channel	median $\mathcal{B}$	max $\mathcal{B}$	over the limit
SM-like	$\mu\tau$	1.7×10^{-4}	0.14	1078/5919 (18%)
SM-like	$e\tau$	1.2×10^{-33}	1.6×10^{-31}	0/5919
SM-like	$e\mu$	4.4×10^{-34}	9.6×10^{-32}	0/5919
h_0 (gauge-phobic)	$\mu\tau$	3.9×10^{-33}	6.2×10^{-30}	0/5919
h_0 (gauge-phobic)	$e\tau$	5.5×10^{-3}	3.2*	5652/5919*
h_0 (gauge-phobic)	$e\mu$	5.0×10^{-2}	11.9*	5919/5919*

CMS limits [3, 4]: $\mathcal{B}(h \to \mu\tau) < 0.15\%$, $\mathcal{B}(h \to e\tau) < 0.22\%$, $\mathcal{B}(h \to e\mu) < 4.4 \times 10^{-5}$ at 95% CL. *For h_0 the "branching ratio" is a ratio to the SM-like width, not its own (open question 1), and the limits are for the 125 GeV state, which h_0 is not. These are rate ratios, not exclusions; the applicable comparison is [4]'s search for additional $e\mu$ resonances at 110–160 GeV, which needs h_0's production cross section.

$h \to \mu\tau$ bounds μ_3^ℓ, not the scalar sector. The SM-like over-the-limit fraction is carried almost entirely by μ_3^ℓ:

μ_3^ℓ [GeV]	0.5	0.9	1.4
SM-like $\mathcal{B}(h \to \mu\tau)$, median	3.9×10^{-5}	2.5×10^{-4}	1.4×10^{-3}
fraction above 0.15%	0.3%	6.4%	48%

The median point over the full scan sits a factor of about 9 *below* the limit. (An earlier version of this result said the model "generically overshoots". That was computed with the wrong δ *and* before the coupling cut on the scalar scan, which together moved the SM-like median from 1.9×10^{-3} to 1.7×10^{-4}.)

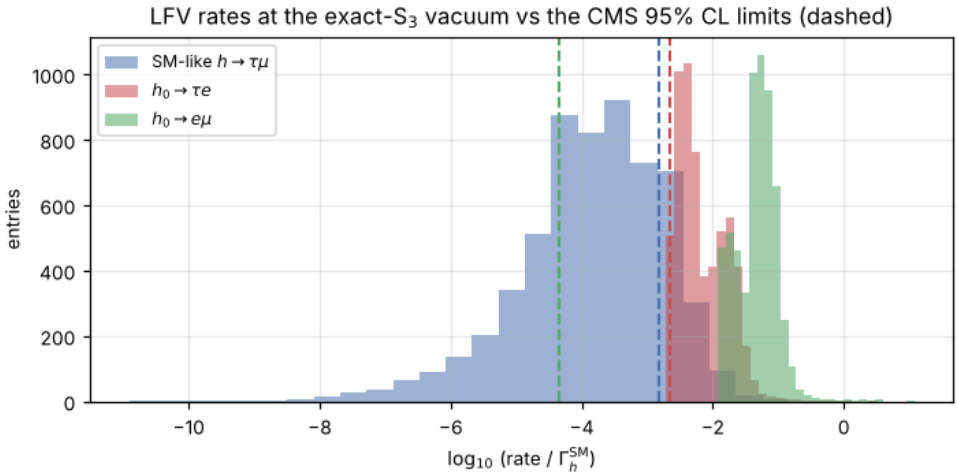


Figure 2: LFV branching ratios of the SM-like state over all points and μ_3^ℓ values, and of h_0's $e\tau$, $e\mu$ rate ratios, against the CMS 95% CL limits (dashed). The SM-like $h \to \tau\mu$ straddles its limit.

The flavour pattern is structural. The largest SM-like $e\tau$ or $e\mu$ value, and the largest h_0 $\mu\tau$ value, anywhere in the scan are below 10^{-29}. That is not a small number, it is zero. O_{12} separates

the electron exactly along the vacuum direction, which is the same first-generation decoupling that forces $V_{us} = 0$ in the quark sector (companion fermion report). The exact-S_3 vacuum ties the quark and lepton flavour structures together.

7 Two library gaps closed

VSS decays. $A_i \to Z h_j$ and $H_i^\pm \to W^\pm h_j$ go through derivative couplings $gV^\mu(S\partial_\mu S' - S'\partial_\mu S)$, which the decay engine used to refuse. 1955 of the 1973 benchmark points (99%) have such a channel open, and they are gauge-strength, so omitting them would have made the heavy-state branching ratios wrong, not just incomplete. They are now squared:

$$\Gamma(S \to V S') = \frac{|c|^2\, \lambda^{3/2}(m_S^2, m_V^2, m_{S'}^2)}{16\pi\, m_S^3\, m_V^2},$$

verified against that closed form and against an independent explicit polarization sum. The model has three $A \to Zh$ vertices ($A_1 Z h_0$, $A_2 Z h_1$, $A_2 Z h_2$) and three $H^\pm \to W^\pm h$ vertices with the same pattern.

The spin-0 $\gamma\gamma$ form factor. A 3HDM has two charged Higgses in the $h \to \gamma\gamma$ loop, which needs

$$A_0(\tau) = \frac{\tau - f(\tau)}{\tau^2}, \qquad \tau = \frac{m_h^2}{4m^2}.$$

Its sign is not copied but forced. [5] Eq. (28) gives $F_W \to +7$, $F_t \to -4/3$, $F^+ \to +1/3$. Since the library has $A_1 = -F_W$ and $A_{1/2} = -F_t$ [6], consistency requires $A_0 = -F^+ \to -1/3$. All three form factors reach their decoupling limits ($+4/3$, -7, $-1/3$), and the SM width comes out at $\Gamma(h \to \gamma\gamma) = 9.181$ keV against PDG's ≈ 9.3 keV.

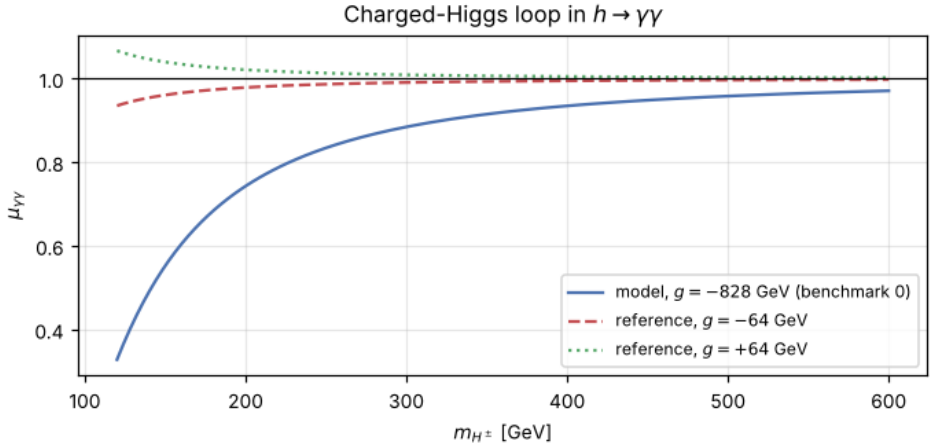


Figure 3: The charged-Higgs contribution to $\mu_{\gamma\gamma}$ against $m_{H^\pm}$. Solid: this model's largest trilinear at benchmark point 0, $g(h_1 H_1^+ H_1^-) = -828$ GeV. Dashed and dotted: reference couplings ∓ 64 GeV. The loop decouples as $m_{H^\pm} \to \infty$, as it must.

8 Open questions, and what would help

1. **The h_0 "branching ratios" are ratios to the SM-like total width.** That is the only width this build has, and it is not h_0's own. These numbers are rate ratios, not branching fractions: 29 of the 17757 h_0 entries (all three channels) exceed 1, which shows the artefact rather than a new problem. The SM-like numbers are unaffected.
2. **The quark channels are absent**, so the total width is the SM prediction rather than one computed from the model, and there are no $b\bar{b}/c\bar{c}$ branching ratios. This is deliberate: the exact-S_3 quark sector is excluded ($V_{us} = 0$), and fitting it would mean quoting rates from a sector known to be wrong. The heavy-state branching ratios ($A_{1,2}$, $H^{\pm}$) need the same quark input to be complete.
3. **The rates in the soft-broken vacuum** are now done, regenerated from the full 3×3 mixing (companion soft-vacuum decays report). The quark channels are still missing there.
4. $Z\gamma$ would need a two-argument charged-scalar form factor, the analogue of A_0. Untouched.
5. μ_3^{ℓ} is scanned rather than fixed. Is there a principled way to fix it, for example from neutrino-sector input, or should the LFV bound on it be the quoted result?

A The derivation ledger

The full typeset version is the companion document `derivations_03.tex`; in summary:

claim	status	independent checks
$g_{h_i VV} = g^{\mathrm{SM}}(\cos\delta, 0, \sin\delta)$, with the sum rule	derived	R_A's first column verified to be $\hat{n}$; SM reference read off the $\delta \to 0$ limit; W and Z agree; the zero survives a generic vacuum angle
The draft's Scenario-C relations hold exactly, for all δ	derived	symbolic identity at general φ, θ_v, δ and symbolic G_k
lepton_mus(μ_3) inverts the mass relations	derived	singular values match the PDG masses to 10^{-9} across the whole μ_3 scan; the Yukawas genuinely vary; a 1% change in μ_4 breaks it
At the exact-S_3 vacuum $h_{1,2}$ give only $\tau\mu$, h_0 only $e\tau$, $e\mu$	derived	derived from the reflection structure of the doublet Yukawa blocks; SM-like $\mathcal{B}(\tau e)$, $\mathcal{B}(e\mu)$ and h_0 $\mathcal{B}(\tau\mu)$ below 10^{-20} at all 35 514 entries
$A_{1/2}$, A_1, A_0 and the LFV width	**imported** [6, 5]	decoupling limits 4/3, -7, $-1/3$; the SM diphoton width is reproduced

References

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[3] CMS Collaboration, *Search for lepton-flavor violating decays of the Higgs boson in the $\mu\tau$ and $e\tau$ final states in pp collisions at $\sqrt{s} = 13$ TeV*, Phys. Rev. D **104**, 032013 (2021), arXiv:2105.03007, doi:10.1103/PhysRevD.104.032013.

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[8] S. Navas *et al.* (Particle Data Group), *Review of Particle Physics*, Phys. Rev. D **110**, 030001 (2024), doi:10.1103/PhysRevD.110.030001.